

SKILL WEEK 13

Task 1:

Code:

```
#include <stdio.h>

int add(int a, int b) {
    return a + b;
}

int multiply(int a, int b) {
    return a * b;
}

int main() {
    int a = 5, b = 3;
    printf("Module integration test\n");
    printf("add(%d,%d) = %d\n", a, b, add(a,b));
    printf("multiply(%d,%d) = %d\n", a, b, multiply(a,b));
    printf("End-to-end execution successful.\n");
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 13.1.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 13.1.c -o 13.1
ashishreddy@Ashishs-MacBook-Air ~ % ./13.1
Module integration test
add(5,3) = 8
multiply(5,3) = 15
End-to-end execution successful.
```

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Task 2:

Code:

```
#include <stdio.h>
#include <errno.h>
#include <string.h>

int main() {
    FILE *fp = fopen("missing.txt", "r");

    if (fp == NULL) {
        fprintf(stderr, "Runtime error: %s\n", strerror(errno));
        fprintf(stderr, "Recovery: check the file path and try again.\n");
        return 1;
    }

    fclose(fp);
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 13.2.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 13.2.c -o 13.2
ashishreddy@Ashishs-MacBook-Air ~ % ./13.2
Runtime error: No such file or directory
Recovery: check the file path and try again.
```

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Task 3:

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *data = malloc(5 * sizeof(int));

    if (data == NULL)
        return 1;

    for (int i = 0; i < 5; i++)
        data[i] = i + 1;

    printf("Allocated values: ");
    for (int i = 0; i < 5; i++)
        printf("%d ", data[i]);
    printf("\n");

    free(data);
    printf("Memory released successfully.\n");

    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 13.3.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc -g 13.3.c -o 13.3
ashishreddy@Ashishs-MacBook-Air ~ % valgrind --leak-check=full ./13.3
Allocated values: 1 2 3 4 5
Memory released successfully.
All heap blocks were freed -- no leaks are possible.
```

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Task 4:

Code:

```
#!/bin/bash

gcc 13.4.c -o 13.4 || exit 1

output=$(./13.4)
expected="PASS"

if [ "$output" = "$expected" ]; then
    echo "Test 1: PASS"
else
    echo "Test 1: FAIL"
fi

echo "Test execution completed."
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 13.4.sh
ashishreddy@Ashishs-MacBook-Air ~ % chmod +x 13.4.sh
ashishreddy@Ashishs-MacBook-Air ~ % ./13.4.sh
Test 1: PASS
Test execution completed.
```